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# SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

**Symbiosis International (Deemed University)**

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

## Assignment No. 01

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|---|---|
| **Subject:** | DevOps Lab |
| **TCODE:** | TE7950 |
| **Name of Student** | **Raghav Sharma** |
| **PRN No.** | **23070122171** |
| **Branch** | CSE, Batch (2023-27), L3 |
| **Academic Year & Semester** | 25 – 26, Sem VII |
| **Date of Performance** | 23rd July, 2026 |
| **Title of Assignment:** | Git Setup & Basic Workflow |
| **Theory:** | Git is a distributed version control system used to track changes in source code during software development. It enables developers to manage project versions, collaborate efficiently, and maintain a complete history of modifications. Basic Git operations include repository initialization, staging, committing, and viewing project status. |
| **Implementation:** | Step 1: Open Git Bash and create a new Git repository using the git init command.<br>`C:\Users\sharm\Desktop>git init`<br>`Initialized empty Git repository in C:/Users/sharm/Desktop/.git/`<br>Step 2: Create a sample file and check the repository status using git status. |

```
C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        gitPractice/
        readme.md.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Step 3: Stage the file using the git add command.

```
C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git add .

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   Readme.md
        new file:   TemporaryReadme.txt
```

Step 4: Commit the staged changes using git commit -m "Initial Commit".

```
C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git commit -m "First Commit"
[master (root-commit) 4a0478f] First Commit
 2 files changed, 1 insertion(+)
 create mode 100644 Readme.md
 create mode 100644 TemporaryReadme.txt

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git status
On branch master
nothing to commit, working tree clean

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git remote

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git remote add origin https://github.com/Gunveer06/gitPractice.git
```

Step 5: Create a new branch using git branch and switch to it.

```
C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git branch feature-update

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git checkout feature-update
Switched to branch 'feature-update'

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git branch
* feature-update
  master

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git status
On branch feature-update
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Readme.md

no changes added to commit (use "git add" and/or "git commit -a")
```

Step 6: Make changes in the new branch and commit the changes.

```
C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git add .

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git commit -m "First Commit to this branch"
[feature-update a3fe4d5] First Commit to this branch
 1 file changed, 1 insertion(+)
```

Step 7: Merge the newly created branch into the main branch.

```
C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git branch
  feature-update
* master

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git status
On branch master
nothing to commit, working tree clean

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git merge feature-update
Updating 4a0478f..a3fe4d5
Fast-forward
 Readme.md | 1 +
 1 file changed, 1 insertion(+)

C:\Users\Admin\Desktop\sem7\Devops Lab\Task1>git status
On branch master
nothing to commit, working tree clean
```

Subject Faculty Name: Prof. Arijit Dutta